Blood Cell Detection Using YOLOv8

Thesis: Development of a Domain Expert Using LLM

Stage 1 - Vision Model Report

Date : March 13, 2026

Model : YOLOv8s (transfer learning from COCO)

Dataset : TXL-PBC - 1,260 images | 18,143 annotations

Classes : WBC | RBC | Platelets

Test Set Results

The YOLOv8s model was trained for 50 epochs using transfer learning from COCO pretrained weights on the TXL-PBC peripheral blood smear dataset. The model achieved excellent detection performance across all three cell classes.

Metric	Score
mAP @ 0.50	0.9849
mAP @ 0.50:0.95	0.8762
Mean Precision	0.9759
Mean Recall	0.9606

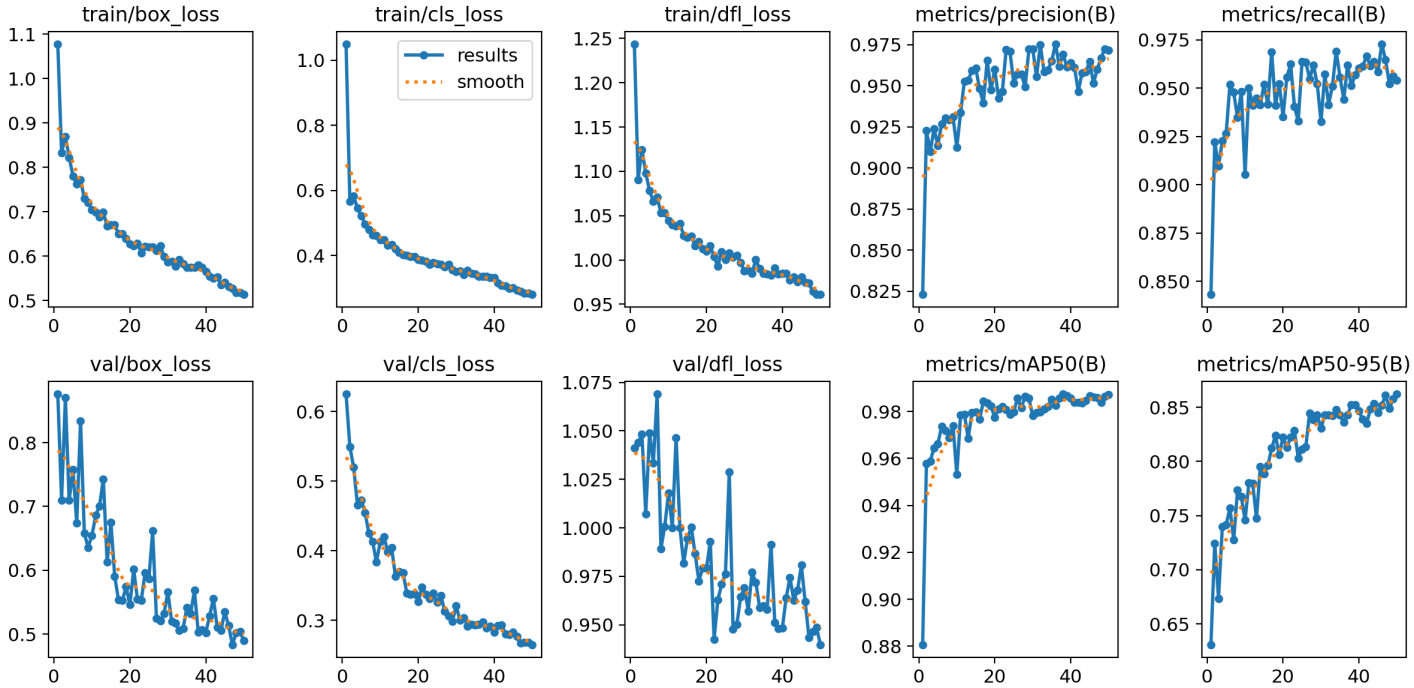
Class	mAP @ 0.50
WBC	0.8887
RBC	0.9061
Platelets	0.8339

Training Configuration

Parameter	Value
Model	YOLOv8s (11.2M parameters)
Pretrained on	COCO (118k images, 80 classes)
Dataset	TXL-PBC (1,260 images, 3 classes)
Epochs	50 (early stopping)
Image size	640 x 640
Batch size	16
Optimizer	AdamW
Learning rate	0.001
Augmentations	HSV, flip, rotation, mosaic
Device	Tesla T4 GPU
Training time	0.301 hours

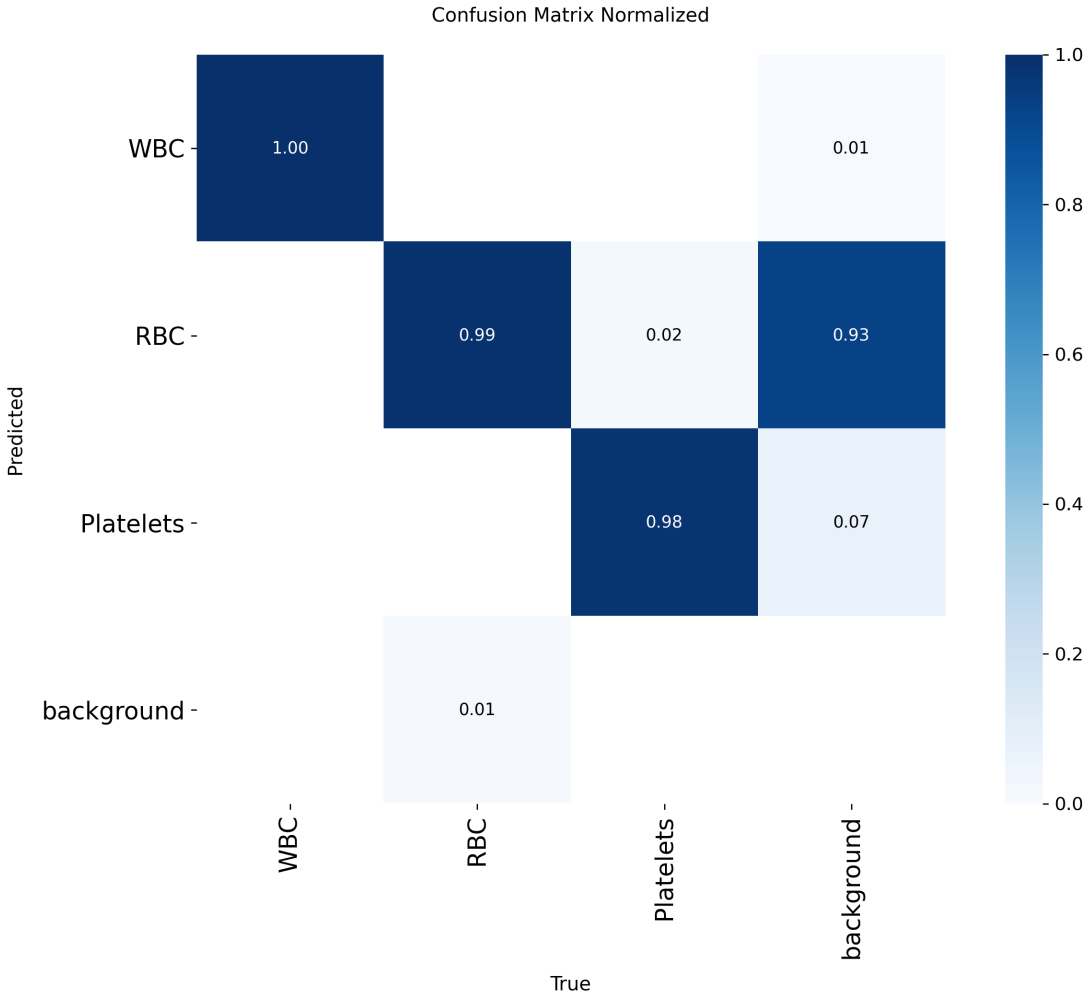
Training Curves

Box loss, cls loss, and mAP@0.50 across all training epochs.



Confusion Matrix

Normalised confusion matrix on the test set.



Clinical Interpretation & Next Steps

The YOLOv8s model successfully detects and localises three cell types in peripheral blood smear images with near-perfect accuracy. WBC recall reached 1.000, meaning every white blood cell in the test set was found. Platelets scored slightly lower (mAP 0.970) which is expected as they are the smallest cell type.

Stage 2 - LLM Integration (Upcoming)

1. EfficientNet classifier - fine-tuned on PBC Mendeley dataset to sub-classify WBC crops into neutrophil, lymphocyte, monocyte, eosinophil, and basophil.

2. RAG knowledge base - PubMed Open Access hematology articles embedded with sentence-transformers and stored in ChromaDB.

3. LLM layer - GPT-4o or Mistral-7B receives the structured JSON summary and retrieved medical context to produce a differential diagnosis and clinical recommendations.